
REPRODUCING LLMSYN – SYNTHETIC EHR GENERATION WITHOUT PATIENT DATA

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Big Data For Healthcare

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INTRODUCTION & GOAL

- **Problem:** Access to real EHRs is limited by HIPAA/GDPR and re-identification risks.
- **Solution (LLMSYN):** Uses large language models (LLMs) to generate realistic synthetic EHRs without training on patient-level data.
- **Method:** 4-step Markov-style prompting pipeline



- **Goal of our project:**
 - Reproduce the LLMSYN pipeline using MIMIC-III data
 - Evaluate Utility (predictive power) and Fidelity (statistical realism)

PROMPT DESIGN & PIPELINE IMPLEMENTATION

Four prompt templates created:

- **Step 1 – Demographics:** Generates age, language, religion, marital status, ethnicity, insurance, mortality flag using priors.
- **Step 2 – Main Diagnosis:** Chooses ICD-9 code using top-100 frequency list + demographics.
- **Step 3 – Comorbidities:** Adds 0–6 secondary ICD-9 codes using Mayo Clinic text.
- **Step 4 – Procedures:** Generates ICD9 procedure codes consistent with previous steps.

Technique: Chain-of-Thought + In-Context Learning → structured tabular output per step.

Markov Dependency: Each stage consumes previous outputs.

Runtime: ~57 s per prompt; 4–6 prompts / patient; 100 patients $\approx$ 4 minutes batch.

Validation: Statistical checks (KS/MMD).

EVALUATION SETUP

Utility

Tasks: Respiratory phenotyping
In-hospital mortality

Training regimes: REAL, FS, DA

Model: Random Forest (200 trees)

Metrics: Accuracy, AUROC

Fidelity

Tasks: Distribution comparison

Metrics: Kolmogorov–Smirnov (KS),
Maximum Mean Discrepancy (MMD)

UTILITY RESULTS (PHENOTYPING & MORTALITY)

Phenotyping Prediction

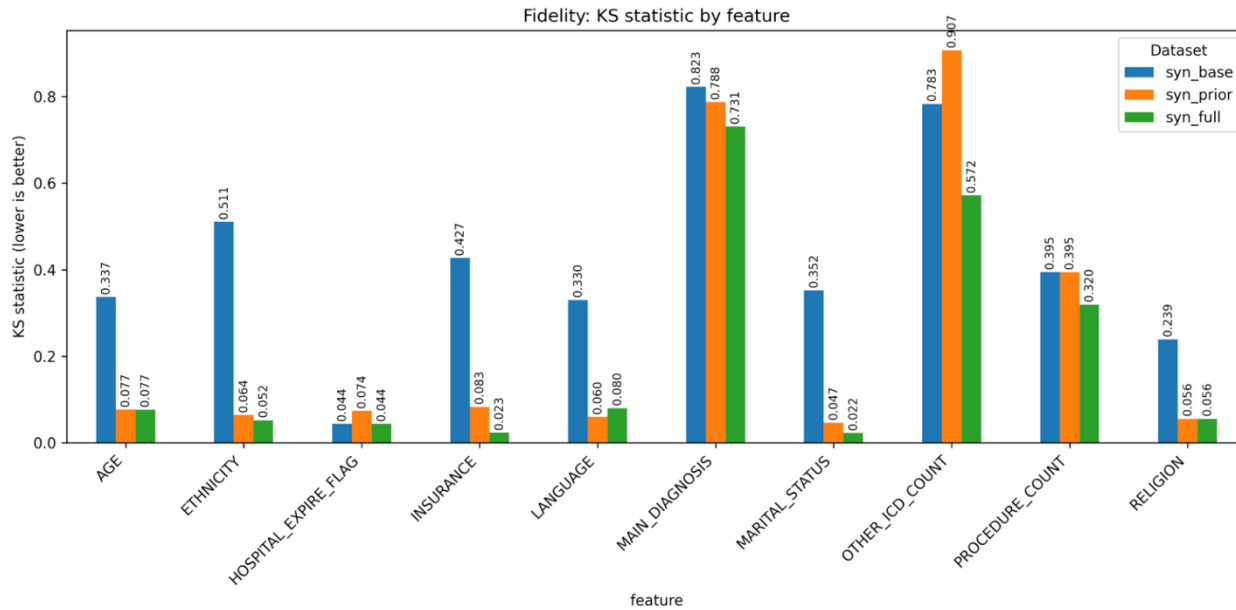
Regime	Variant	Train	Test	ACC	AUROC
DA	syn base	REAL(100)+SYN(100)	REAL(100)	0.960 ± 0.000	0.512 ± 0.153
DA	syn full	REAL(100)+SYN(100)		0.960 ± 0.000	0.540 ± 0.184
DA	syn prior	REAL(100)+SYN(100)		0.960 ± 0.000	0.534 ± 0.175
FS	syn base	SYN(100)		0.960 ± 0.000	0.452 ± 0.177
FS	syn full	SYN(100)		0.958 ± 0.004	0.624 ± 0.127
FS	syn prior	SYN(100)		0.960 ± 0.000	0.499 ± 0.144
REAL	match	REAL(100)		0.958 ± 0.004	0.528 ± 0.164

- DA $\approx$ or $>$ REAL (matches LLMSYN claim)
- FS usable but unstable
- High Accuracy due to imbalance
→ AUROC is reliable metric
- Prior variant noisy at 100-sample scale

Mortality Prediction

Regime	Variant	Train	Test	ACC	AUROC
DA	syn base	REAL(100)+SYN(100)	REAL(100)	0.885 ± 0.008	0.590 ± 0.091
DA	syn full	REAL(100)+SYN(100)		0.881 ± 0.007	0.607 ± 0.112
DA	syn prior	REAL(100)+SYN(100)		0.882 ± 0.010	0.573 ± 0.114
FS	syn base	SYN(100)		0.867 ± 0.013	0.506 ± 0.060
FS	syn full	SYN(100)		0.831 ± 0.043	0.516 ± 0.079
FS	syn prior	SYN(100)		0.866 ± 0.014	0.425 ± 0.108
REAL	match	REAL(100)		0.877 ± 0.016	0.602 ± 0.108

FIDELITY RESULTS (KS & MMD)



MMD² between synthetic and real joint distributions (lower = better).

Pair	Full	Prior	Base
MAIN + PROCEDURE	0.0385	0.0260	0.0259
MAIN + COMORBIDITY	0.1770	0.1601	0.1596
COMORB + PROCEDURE	0.1842	0.0184	0.0181

KS

- **Demographics fidelity strong** (KS 0.02–0.08)
- **Diagnosis/comorbidity fidelity poor** (KS > 0.7)

MMD

- **Joint structure hardest**
(COMORB+PROC worst)
- **Base/Prior sometimes closer**
than Full (small-sample effect)

ABLATION: REMOVING COMORBIDITIES

- Set **OTHER_ICD_COUNT = 0**
- Utility unchanged (AUROC similar)
- Comorbidity KS = 0, but **MAIN+COMORB still high**
- **COMORB+PROC worsened**
 - Comorbidities add richness, but **little utility impact** at 100 samples

DISCUSSION & KEY TAKEAWAYS

- LLMSYN is **partially reproducible**
 - Qualitative trends reproduced
 - $DA \geq REAL$
 - $FS < REAL$ but usable
 - Strong demographics fidelity
 - **Not reproducible:** diagnosis & comorbidity fidelity
 - **Reasons:**
 - Only 100 synthetic patients
 - Different LLMs & priors
 - Missing original prompts + Mayo grounding
- Core idea holds, but **clinical structure remains main bottleneck**